

## TECHNICAL DATA SHEET

## HD T9

**HD T9** is a HDPE stretched tape / raffia grade produced by Spherilene Technology

This grade can be used for stretched tape / raffia for woven fabrics & tarpaulin

**HD T9** is characterized by excellent processability and mechanical strength

**BIS Designation Code:** IS 7328-3B-TB-EDA

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.1           |
| Melt Flow Index (5 kg, 190°C)          |                                         |                   | 3             |
| Melt Flow Index (21.6 kg, 190°C)       | ASTM D1238                              | g/10 min          | 27            |
| Density (23°C)                         | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 26            |
| Tensile Strength at Break              |                                         | MPa               | 38            |
| Elongation at Yield                    |                                         | %                 | 11            |
| Elongation at Break                    |                                         | %                 | 1100          |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 400           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 1100          |
| Vicat Softening Point (10N)            | ASTM D1525                              | °C                | 127           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 133           |
| Hardness                               | ASTM D2240                              | Shore D           | 61            |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 240 °C                            |                   |               |
| Die Temperature                        | 230 – 250 °C                            |                   |               |
| Quench Temperature                     | 30 °C                                   |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## HD T10

**HD T10** is a HDPE yarn / monofilament grade produced by Spherilene Technology

This grade can be used in yarns / monofilament for Fabrics, Nets, Twines & Ropes

**HD T10** is characterized by excellent processability and mechanical strength

**BIS Designation Code:** IS 7328-3B-LBO-EDA

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.1           |
| Melt Flow Index (5 kg, 190°C)          |                                         |                   | 3             |
| Melt Flow Index (21.6 kg, 190°C)       | ASTM D1238                              | g/10 min          | 27            |
| Density (23°C)                         | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 26            |
| Tensile Strength at Break              |                                         | MPa               | 38            |
| Elongation at Yield                    |                                         | %                 | 11            |
| Elongation at Break                    |                                         | %                 | 1100          |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 400           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 1000          |
| Vicat Softening Point (10N)            | ASTM D1525                              | °C                | 127           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 133           |
| Hardness                               | ASTM D2240                              | Shore D           | 61            |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 240 °C                            |                   |               |
| Die Temperature                        | 230 – 250 °C                            |                   |               |
| Quench Temperature                     | 30 °C                                   |                   |               |
| Stretching Temperature                 | 95 – 120 °C                             |                   |               |

\* Halene H is the registered trademark of High Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## HD T10S

**HD T10S** is a HDPE monofilament grade produced by Spherilene Technology

This grade can be used in monofilament for Nets, Twines & Ropes

**HD T10S** is characterized by excellent processability and mechanical strength

**BIS Designation Code:** IS 7328-3B-LB-EDA

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.1           |
| Melt Flow Index (5 kg, 190°C)          |                                         |                   | 3             |
| Melt Flow Index (21.6 kg, 190°C)       |                                         |                   | 27            |
| Density (23°C)                         | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 26            |
| Tensile Strength at Break              |                                         | MPa               | 38            |
| Elongation at Yield                    |                                         | %                 | 11            |
| Elongation at Break                    |                                         | %                 | 1100          |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 400           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 1000          |
| Vicat Softening Point (10N)            | ASTM D1525                              | °C                | 127           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 131           |
| Hardness                               | ASTM D2240                              | Shore D           | 61            |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 240 °C                            |                   |               |
| Die Temperature                        | 230 – 250 °C                            |                   |               |
| Quench Temperature                     | 30 °C                                   |                   |               |
| Stretching Temperature                 | ~ 98 °C (Water)                         |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## M5002L

**M5002L** is a Narrow MWD HDPE produced by Spherilene Technology

**M5002L** is suited for injection and extrusion-compression molded Caps & Closures

The precise control of morphology & MWD during polymerization confers superior processability, excellent organoleptics & improved performance properties compared to conventional HDPE grades

**BIS Designation Code:** IS 7328-3B-CB-EDB

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 3             |
| Melt Flow Index (5 kg, 190°C)           |                                         |                   | 8             |
| Melt Flow Index (21.6 kg, 190°C)        | ASTM D1238                              | g/10 min          | 70            |
| Density (23°C)                          | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Tensile Strength at Break               |                                         | MPa               | 30            |
| Elongation at Break                     |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 180           |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 950           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 70            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 125           |
| DSC Melting Temperature                 | ASTM D3418                              | °C                | 132           |
| Hardness                                | ASTM D2240                              | Shore D           | 61            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 160 – 190 °C                            |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## M5005L

**M5005L** is a Narrow MWD HDPE produced by Spherilene Technology

**M5005L** is recommended for injection molding of heavy-duty Crates, Industrial products and for injection molded as well as extrusion compression molded caps for packaged drinking water

The precise control of morphology & MWD during polymerization confers superior processability & improved performance properties compared to conventional HDPE grades

**BIS Designation Code:** IS 7328-3B-MB-EDC

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 4.5           |
| Melt Flow Index (5 kg, 190°C)           |                                         |                   | 10            |
| Melt Flow Index (21.6 kg, 190°C)        | ASTM D1238                              | g/10 min          | 110           |
| Density (23°C)                          | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 22            |
| Tensile Strength at Break               |                                         | MPa               | 31            |
| Elongation at Break                     |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 100           |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 850           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 70            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 124           |
| DSC Melting Temperature                 | ASTM D3418                              | °C                | 132           |
| Hardness                                | ASTM D2240                              | Shore D           | 62            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 185 – 235 °C                            |                   |               |
| Nozzle Temperature                      | 235 °C                                  |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## M5018L

**M5018L** is an Injection Molding HDPE produced by Spherilene Technology

**M5018L** is recommended for general purpose Injection Molding

**M5018L** combines excellent processability with good mechanical properties

**BIS Designation Code:** IS 7328-3B-MB-EDE

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 20            |
| Melt Flow Index (5 kg, 190°C)           |                                         |                   | 50            |
| Melt Flow Index (21.6 kg, 190°C)        | ASTM D1238                              | g/10 min          | 450           |
| Density (23°C)                          | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 22            |
| Elongation at yield                     |                                         | %                 | 10            |
| Elongation at Break                     |                                         | %                 | 450           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 35            |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 800           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 64            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 121           |
| DSC Melting Temperature                 | ASTM D3418                              | °C                | 132           |
| Hardness                                | ASTM D2240                              | Shore D           | 62            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 180 – 235 °C                            |                   |               |
| Nozzle Temperature                      | 235 °C                                  |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test Specimens

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## M5025L

**M5025L** is an Injection Molding HDPE produced by Spherilene Technology

This resin is particularly recommended for general purpose Injection Molded items & Masterbatch application

**M5025L** combines excellent processability with good mechanical properties

| Property                                | Test Method | Unit              | Value     |
|-----------------------------------------|-------------|-------------------|-----------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238  | g/10 min          | 30.0      |
| Density (23°C)                          | ASTM D1505  | g/cm <sup>3</sup> | 0.950     |
| <b>Physical Property</b>                |             |                   |           |
| Tensile Strength at Yield               | ASTM D638   | MPa               | 20        |
| Elongation at Break                     | (50 mm/min) | %                 | 250       |
| Notched Izod Impact Strength (23°C)     | ASTM D256A  | J/m               | 28        |
| Flexural Modulus                        | ASTM D790A  | MPa               | 800       |
| Hardness                                | ASTM D2240  | Shore D           | 62        |
| Spiral Flow Length @ 235°C              | HPL Method  | mm                | 730       |
| Vicat Softening Point (10 N)            | ASTM D1525  | °C                | 125       |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648   | °C                | 77        |
| DSC Melting Temperature                 | ASTM D3418  | °C                | 132       |
| <b>Suggested Processing Conditions</b>  |             |                   |           |
| Barrel Temperature                      |             | °C                | 180 - 235 |
| Nozzle Temperature                      |             | °C                | 235       |

\* Halene H is the registered trademark of High Density Polyethylene of HPL

Mechanical Properties are on Injection Molded test specimens prepared as per ASTM D4101

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## TECHNICAL DATA SHEET

## M6007L

**M6007L** is a Narrow MWD HDPE produced by Spherilene Technology

**M6007L** is ideally suited for injection molding of heavy-duty Crates & Industrial products.

The precise control of morphology & MWD during polymerization confers superior processability & improved performance properties compared to conventional HDPE grades

**BIS Designation Code:** IS 7328-3B-MB-EDD

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 7.5           |
| Melt Flow Index (5 kg, 190°C)           |                                         |                   | 22            |
| Melt Flow Index (21.6 kg, 190°C)        | ASTM D1238                              | g/10 min          | 190           |
| Density (23°C)                          | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 24            |
| Elongation at Break                     |                                         | %                 | 750           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 50            |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 850           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 67            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 122           |
| DSC Melting Temperature                 | ASTM D3418                              | °C                | 132           |
| Hardness                                | ASTM D2240                              | Shore D           | 62            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 185 – 235 °C                            |                   |               |
| Nozzle Temperature                      | 200 – 235 °C                            |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

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## TECHNICAL DATA SHEET

## M6007LU

**M6007LU** is a Narrow MWD HDPE produced by Spherilene Technology

**M6007LU** is ideally suited for injection molding of heavy-duty Crates & Industrial products

**M6007LU** is stabilized adequately with HALS\* to give long service life under outdoor exposure

The precise control of morphology & MWD during polymerization confers superior processability & improved performance properties compared to conventional HDPE grades

**BIS Designation Code:** IS 7328-3B-MD-EDD

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)        | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 7.5           |
| Melt Flow Index (5 kg, 190°C)           |                                         |                   | 22            |
| Melt Flow Index (21.6 kg, 190°C)        | ASTM D1238                              | g/10 min          | 190           |
| Density (23°C)                          | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 24            |
| Elongation at Break                     |                                         | %                 | 800           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 50            |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 850           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 67            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 122           |
| DSC Melting Temperature                 | ASTM D3418                              | °C                | 132           |
| Hardness                                | ASTM D2240                              | Shore D           | 62            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 185 – 235 °C                            |                   |               |
| Nozzle Temperature                      | 200 – 235 °C                            |                   |               |

\* Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens

+ The total Hindered Amine Light Stabilizer (HALS) content is ~ 2000 ppm

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## B5500

**B5500** is a HMHDPE Blow Molding Grade produced by Mitsui CX Process

**B5500** combines exceptional processability with high rigidity, excellent mechanical properties and impact resistance

**B5500** is recommended for medium bulk containers (~ 120 liter) for Chemicals, Detergent, Lube oils

**B5500** is also suitable for the manufacture of blow molded water tanks up to 1000 liters. Adequate UV Stabilizer needs to be added externally for usage in outdoor water tank applications

**BIS Designation Code:** IS 7328-3B-BB-FXTA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.06          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 0.35          |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 8             |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.952         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.956         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 27            |
| Elongation at Yield                           |                                         | %                 | 9             |
| Tensile Strength at Break                     |                                         | MPa               | 40            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 500           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| Hardness                                      | ASTM D2240                              | Shore D           | 64            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >1000         |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 127           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 63            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 133           |

| Suggested Processing Conditions |                    |
|---------------------------------|--------------------|
| Barrel Temperature              | 200/210/215/210 °C |
| Die Temperature                 | 200 – 210 °C       |
| Parison Temperature             | 190 – 210 °C       |
| Cooling Water for Mold          | 20 – 25 °C         |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## B6401

**B6401** is a HDPE Blow Molding Grade produced by Mitsui CX Process

**B6401** combines exceptional processability with high rigidity, excellent mechanical properties and impact resistance

**B6401** is recommended for upto 20 L containers for Lubricants and Edible oils

**BIS Designation Code:** IS 7328-3P-BB-GXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.40          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 1.6           |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 25            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.960         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.963         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 32            |
| Elongation at Yield                           |                                         | %                 | 9             |
| Tensile Strength at Break                     |                                         | MPa               | 40            |
| Elongation at Break                           |                                         | %                 | 1000          |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 300           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1300          |
| Hardness                                      | ASTM D2240                              | Shore D           | 68            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | < 20          |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 129           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 70            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 135           |

| Suggested Processing Conditions |                    |
|---------------------------------|--------------------|
| Barrel Temperature              | 150/160/165/175 °C |
| Die Temperature                 | 180 – 185 °C       |
| Parison Temperature             | 185 – 190 °C       |
| Cooling Water for Mold          | 20 – 25 °C         |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## E5201

**E5201** is a broad molecular weight distribution HDPE Grade produced by Mitsui CX Process

**E5201** combines exceptional processability with high ESCR and excellent mechanical properties

**E5201** is particularly recommended for High ESCR Blow Molding, Cable Sheathing & Jacketing Compound, Lamination Film, General Purpose Film and Extrusion Applications

**BIS Designation Code:** IS 7328-3B-BB-EXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.35          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 1.6           |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 32            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.952         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Elongation at Yield                           |                                         | %                 | 7             |
| Tensile Strength at Break                     |                                         | MPa               | 34            |
| Elongation at Break                           |                                         | %                 | 850           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 120           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >600          |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 122           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 60            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 130           |

| Suggested Processing Conditions |              |
|---------------------------------|--------------|
| Barrel Temperature              | 150 – 160 °C |
| Die Temperature                 | 160 – 165 °C |
| Parison Temperature             | 160 – 170 °C |
| Cooling Water for Mold          | 20 – 25 °C   |

\*Halene H is the registered trademark of High Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are tested on specimens from Compression Molded sheets

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

## E5201S

**E5201S** is a broad molecular weight distribution HDPE Grade produced by Mitsui CX Process

**E5201S** combines exceptional processability and excellent mechanical properties

**E5201S** is recommended for General Purpose Blow Molding Applications. It can also be used for Double Walled Corrugated Pipes & General-Purpose Pipe Applications

**BIS Designation Code:** IS 7328-3B-BB-FXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.35          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 1.6           |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 32            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.954         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.958         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Elongation at Yield                           |                                         | %                 | 9             |
| Tensile Strength at Break                     |                                         | MPa               | 36            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 100           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1100          |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 250           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 125           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 70            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 130           |

| Suggested Processing Conditions |              |
|---------------------------------|--------------|
| Barrel Temperature              | 150 – 160 °C |
| Die Temperature                 | 160 – 165 °C |
| Parison Temperature             | 165 – 170 °C |
| Cooling Water for Mold          | 20 – 25 °C   |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## F5400

**F5400** is a HMHDPE film grade produced by Mitsui CX Process

**F5400** combines excellent processability and high mechanical strength

**F5400** is recommended for General Purpose Film as well as very thin Reinforced Bag Applications

**BIS Designation Code:** IS 7328-3B-FB-FXTA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.09          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 0.50          |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 14            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.952         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.954         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 28            |
| Tensile Strength at Break                     |                                         | MPa               | 40            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 180           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 950           |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >500          |
| Dart Impact Strength**                        | ASTM D1709A                             | g                 | 80            |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 124           |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 180 – 220 °C                            |                   |               |
| BUR                                           | 4 – 6                                   |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

\*\*Dart Impact Strength tested on 20 micron blown film made with BUR 4.0

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## HD T6

**HD T6** is a HDPE General Purpose Extrusion & Lamination Film Grade produced by Mitsui CX Process

**HD T6** combines good processability with high mechanical strength.

**HD T6** is recommended for Lamination films, Woven sacks, and Tarpaulin.

**BIS Designation Code:** IS 7328-3P-FB-GXDA

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.50          |
| Melt Flow Index (5 kg, 190°C)          |                                         | g/10 min          | 1.8           |
| Melt Flow Index (21.6 kg, 190°C)       |                                         | g/10 min          | 30            |
| Density (23°C, Annealed)               | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.960         |
| Density (23°C, Annealed)               | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.961         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 30            |
| Tensile Strength at Break              |                                         | MPa               | 38            |
| Elongation at Break                    |                                         | %                 | 800           |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 250           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 1150          |
| Hardness                               | ASTM D2240                              | Shore D           | 65            |
| Vicat Softening Point (10 N)           | ASTM D1525                              | °C                | 124           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 240 °C                            |                   |               |
| Die Temperature                        | 220 – 240 °C                            |                   |               |
| Quench Temperature                     | 25 °C                                   |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

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## TECHNICAL DATA SHEET

## M5600S

**M5600S** is a broad molecular weight distribution HDPE Grade produced by Mitsui CX Process

**M5600S** is suited for both injection & compression molded caps & closure application for Carbonated Soft Drink (CSD), water, juice & beverages

The precise control of morphology & broader MWD during polymerization confers superior processability, good organoleptic, high environmental stress crack resistance & improved performance properties. This grade contains slip agent for better torque release.

**BIS Designation Code:** IS 7328-3B-CBF-FXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.7           |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 2.7           |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.954         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.958         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 26            |
| Elongation at Yield                           |                                         | %                 | 9             |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 75            |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 950           |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 750           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 125           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 68            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 160 – 190 °C                            |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## M5601

**M5601** is a broad molecular weight distribution HDPE Grade produced by Mitsui CX Process

**M5601** is suited for both injection & compression molded caps & closure application for Carbonated Soft Drink (CSD), water, juice & beverages

The precise control of morphology & broader MWD during polymerization confers superior processability, good organoleptic, high environmental stress crack resistance & improved performance properties.

This grade does not contain slip agent

**BIS Designation Code:** IS 7328-3B-CB-FXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 1.2           |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 5.0           |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.954         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.958         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 28            |
| Elongation at Yield                           |                                         | %                 | 9.5           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 50            |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 400           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 124           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 67            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 160 – 190 °C                            |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## M5601S

**M5601S** is a broad molecular weight distribution HDPE Grade produced by Mitsui CX Process

**M5601S** is suited for both injection & compression molded caps & closure application for Carbonated Soft Drink (CSD), water, juice & beverages

The precise control of morphology & broader MWD during polymerization confers superior processability, good organoleptic, high environmental stress crack resistance & improved performance properties. This grade contains slip agent for better torque release.

**BIS Designation Code:** IS 7328-3B-CBF-FXDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 1.2           |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 5.0           |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.954         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.958         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 28            |
| Elongation at Yield                           |                                         | %                 | 9.5           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 50            |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 400           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 124           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 67            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 160 – 190 °C                            |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## M5818

**M5818** is an Injection Molding HDPE produced by Mitsui CX Technology

This resin is particularly recommended for general purpose Injection Molded items

**M5818** combines excellent processability with good mechanical properties

| Property                               | Test Method | Unit              | Value     |
|----------------------------------------|-------------|-------------------|-----------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238  | g/10 min          | 19.0      |
| Density (23°C)                         | ASTM D1505  | g/cm <sup>3</sup> | 0.956     |
| <b>Physical Property</b>               |             |                   |           |
| Tensile Strength at Yield              | ASTM D638   | MPa               | 24        |
| Elongation at Break                    | (50 mm/min) | %                 | 350       |
| Notched Izod Impact Strength (23°C)    | ASTM D256A  | J/m               | 36        |
| Hardness                               | ASTM D2240  | Shore D           | 64        |
| DSC Melting Temperature                | ASTM D3418  | °C                | 131       |
| <b>Suggested Processing Conditions</b> |             |                   |           |
| Barrel Temperature                     |             | °C                | 180 – 235 |
| Nozzle Temperature                     |             | °C                | 235       |

\* Halene H is the registered trademark of High Density Polyethylene of HPL

Mechanical Properties are on Injection Molded test specimens prepared as per ASTM D4101

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## TECHNICAL DATA SHEET

## P5100

**P5100** is a HDPE pipe grade produced by Mitsui CX Process

**P5100** combines excellent processability with good creep resistance, ESCR and mechanical properties

**P5100** is recommended for **PE 80** compliant pressure pipes

**BIS Designation Code:** IS 7328-3B-PB-FXTA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.09          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 0.50          |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 14            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.952         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.955         |
| Density (27°C, Annealed)                      | IS 4984                                 | g/cm <sup>3</sup> | 0.948         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Tensile Strength at Break                     |                                         | MPa               | 38            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 200           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >500          |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 125           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 64            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| Oxidative Induction Time                      | ASTM D3895                              | min               | >30           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 160 – 180 °C                            |                   |               |
| Die Temperature                               | 170 – 190 °C                            |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment.*

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## TECHNICAL DATA SHEET

## P5200

**P5200** is a HDPE pipe grade produced by Mitsui CX Process

**P5200** combines excellent processability with good creep resistance, ESCR and mechanical properties

**P5200** is recommended for **PE 63** compliant pressure pipes

**BIS Designation Code:** IS 7328-3B-PB-EXTA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.20          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 0.90          |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 22            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.954         |
| Density (27°C, Annealed)                      | IS 4984                                 | g/cm <sup>3</sup> | 0.948         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Elongation at Yield                           |                                         | %                 | 8             |
| Tensile Strength at Break                     |                                         | MPa               | 38            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 175           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >1000         |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 123           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 64            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| Oxidative Induction Time                      | ASTM D3895                              | min               | >30           |

| Suggested Processing Conditions |              |
|---------------------------------|--------------|
| Barrel Temperature              | 150 – 180 °C |
| Die Temperature                 | 170 – 180 °C |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

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## TECHNICAL DATA SHEET

## P5200UV

**P5200UV** is a HDPE pipe grade produced by Mitsui CX Process

**P5200UV** combines excellent processability with good creep resistance, ESCR and mechanical properties

**P5200UV** is stabilized adequately with HALS<sup>+</sup> to give a long service life under outdoor exposure & recommended for telecom ducts

**BIS Designation Code:** IS 7328-3B-PD-EXTA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.20          |
| Melt Flow Index (5 kg, 190°C)                 |                                         | g/10 min          | 0.90          |
| Melt Flow Index (21.6 kg, 190°C)              |                                         | g/10 min          | 20            |
| Density (23°C, Annealed)                      | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| Density (23°C, Annealed)                      | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.954         |
| Density (27°C, Annealed)                      | IS 4984                                 | g/cm <sup>3</sup> | 0.948         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 25            |
| Elongation at Yield                           |                                         | %                 | 8             |
| Tensile Strength at Break                     |                                         | MPa               | 38            |
| Elongation at Break                           |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 175           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 1000          |
| Hardness                                      | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | >1000         |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 125           |
| Heat Deflection Temperature (0.455 MPa)       | ASTM D648                               | °C                | 64            |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 131           |
| Oxidative Induction Time                      | ASTM D3895                              | min               | >30           |

| Suggested Processing Conditions |              |
|---------------------------------|--------------|
| Barrel Temperature              | 150 – 180 °C |
| Die Temperature                 | 170 – 180 °C |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

\* The total Hindered Amin Light Stabilizer (HALS) content is ~ 2000 ppm.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

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## TECHNICAL DATA SHEET

## P5300

**P5300** is a HDPE pipe grade produced by Mitsui CX Process

**P5300** combines excellent processability with good creep resistance, high ESCR and excellent mechanical properties

**P5300** has MRS 10 MPa according to ISO 9080 and is designated PE 100 according to ISO 12162

**BIS Designation Code:** IS 7328-3B-PBK-EXTA

| Property                                                             | Test Method                             | Unit              | Nominal Value |
|----------------------------------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)                                     | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.05          |
| Melt Flow Index (5 kg, 190°C)                                        |                                         | g/10 min          | 0.25          |
| Melt Flow Index (21.6 kg, 190°C)                                     |                                         | g/10 min          | 9.5           |
| Density (23°C, Annealed)                                             | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.950         |
| Density (23°C, Annealed)                                             | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.954         |
| Density (27°C, Annealed)                                             | IS 4984                                 | g/cm <sup>3</sup> | 0.948         |
| <b>Physical Property</b>                                             |                                         |                   |               |
| Tensile Strength at Yield                                            | ASTM D638<br>(50 mm/min)                | MPa               | 26            |
| Elongation at Yield                                                  |                                         | %                 | 11            |
| Tensile Strength at Break                                            |                                         | MPa               | 40            |
| Elongation at Break                                                  |                                         | %                 | 900           |
| Notched Izod Impact Strength (23°C)                                  | ASTM D256A                              | J/m               | 270           |
| Flexural Modulus                                                     | ASTM D790A                              | MPa               | 1000          |
| Hardness                                                             | ASTM D2240                              | Shore D           | 65            |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v)                        | ASTM D1693B                             | Hr                | >1000         |
| Vicat Softening Point (10 N)                                         | ASTM D1525                              | °C                | 125           |
| Heat Deflection Temperature (0.455 MPa)                              | ASTM D648                               | °C                | 70            |
| DSC Melting Temperature                                              | ASTM D3418                              | °C                | 131           |
| Oxidative Induction Time                                             | ASTM D3895                              | min               | >30           |
| Resistance to Rapid Crack Propagation at 0 °C<br>– Critical Pressure | ISO 13477                               | Bar               | ≥10           |

| Suggested Processing Conditions |              |
|---------------------------------|--------------|
| Barrel Temperature              | 160 – 220 °C |
| Die Temperature                 | 180 – 220 °C |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

Pre drying of the material at ~ 90 °C for ~ 2 hours before processing is recommended.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

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## TECHNICAL DATA SHEET

## R5801

**R5801** is a HDPE grade for stretched tape / raffia application produced by Mitsui CX Process

**R5801** combines excellent processability with high mechanical strength

**R5801** is recommended for Stretched Tape / Raffia for Woven sacks and Tarpaulin

**BIS Designation Code:** IS 7328-3P-TB-GXDA

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 0.50          |
| Melt Flow Index (5 kg, 190°C)          |                                         | g/10 min          | 1.8           |
| Melt Flow Index (21.6 kg, 190°C)       |                                         | g/10 min          | 30            |
| Density (23°C, Annealed)               | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.960         |
| Density (23°C, Annealed)               | JIS MCI HZ-F-109                        | g/cm <sup>3</sup> | 0.961         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 32            |
| Tensile Strength at Break              |                                         | MPa               | 38            |
| Elongation at Break                    |                                         | %                 | 800           |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 200           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 1150          |
| Hardness                               | ASTM D2240                              | Shore D           | 65            |
| Vicat Softening Point (10 N)           | ASTM D1525                              | °C                | 124           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 131           |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 240 °C                            |                   |               |
| Die Temperature                        | 220 – 240 °C                            |                   |               |
| Quench Temperature                     | 25 – 30 °C                              |                   |               |

\*Halene H is the registered trademark of High-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on specimens from Compression Molded sheet prepared in accordance with ASTM D4703

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

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## TECHNICAL DATA SHEET

## 71501S

**71501S** is a LLDPE blown film grade produced by the Spherilene Technology

This grade can be used in General Purpose Packaging Film Applications

**71501S** is additivated with slip and anti-blocking additives for excellent openability and slip properties

**BIS Designation Code:** IS 7328-2B-FBH-BDA

| Property                                                         | Test Method                             | Unit              | Nominal Value |
|------------------------------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)                                 | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.0           |
| Melt Flow Index (5 kg, 190°C)                                    |                                         |                   | 2.6           |
| Melt Flow Index (21.6 kg, 190°C)                                 |                                         |                   | 23            |
| Density (23°C)                                                   | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.922         |
| DSC Melting Temperature                                          | ASTM D3418                              | °C                | 126           |
| <b>Film Property **</b>                                          |                                         |                   |               |
| Tensile Strength at Yield MD                                     | ASTM D882<br>(500 mm/min)               | MPa               | 10            |
| Tensile Strength at Yield TD                                     |                                         | MPa               | 10            |
| Tensile Strength at Break MD                                     |                                         | MPa               | 25            |
| Tensile Strength at Break TD                                     |                                         | MPa               | 16            |
| Elongation at Break MD                                           |                                         | %                 | 650           |
| Elongation at Break TD                                           |                                         | %                 | 750           |
| Elmendorf Tear Strength MD                                       | ASTM D1922                              | g/μ               | 6             |
| Elmendorf Tear Strength TD                                       |                                         | g/μ               | 16            |
| Dart Impact Strength, F <sub>50</sub> (38 mm Dart, 66 cm Height) | ASTM D1709A                             | g/μ               | 4.0           |
| Gloss at 60°                                                     | ASTM D2457                              | GU                | 76            |
| <b>Suggested Processing Conditions</b>                           |                                         |                   |               |
| Barrel Temperature                                               | 180 – 200 °C                            |                   |               |
| Melt Temperature                                                 | 190 – 210 °C                            |                   |               |
| Blow up Ratio                                                    | 2.0 – 3.0                               |                   |               |
| Die Gap                                                          | 2.0 – 2.5 mm                            |                   |               |

\* Halene L is the registered trademark of Linear Low Density Polyethylene of Haldia Petrochemicals Limited

\*\* Typical values with 25 μ film made with 2.5 mm die gap & BUR 2.5

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 71601D

**71601D** is a LLDPE grade produced by the Spherilene Technology

**71601D** is suitably additivated for excellent processability in Drip Lateral Pipes

The grade is characterized by controlled morphology, excellent processability and mechanical properties

**BIS Designation Code:** IS 7328-2B-NB-BDA

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.0           |
| Melt Flow Index (5 kg, 190°C)                 |                                         |                   | 2.8           |
| Melt Flow Index (21.6 kg, 190°C)              | ASTM D1238                              | g/10 min          | 23            |
| Density (23°C)                                | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.920         |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Break                     | ASTM D638<br>(50 mm/min)                | MPa               | 30            |
| Elongation at Break                           |                                         | %                 | 1000          |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 500           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 400           |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 1000          |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 125           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Barrel Temperature                            | 200 – 240 °C                            |                   |               |
| Die Temperature                               | 240 – 250 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on compression Molded test specimens.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 71601W

**71601W** is a LLDPE blown film grade produced by the Spherilene Technology with controlled morphology, excellent processability & mechanical properties

This grade can be used in General Purpose Adhesive Lamination films

**71601W** is without slip or antiblocking additives

**BIS Designation Code:** IS 7328-2B-FBE-BDA

| Property                                                         | Test Method                             | Unit              | Nominal Value |
|------------------------------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)                                 | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 1.0           |
| Melt Flow Index (5 kg, 190°C)                                    |                                         |                   | 2.6           |
| Melt Flow Index (21.6 kg, 190°C)                                 | ASTM D1238                              | g/10 min          | 23            |
| Density (23°C)                                                   | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.920         |
| DSC Melting Temperature                                          | ASTM D3418                              | °C                | 125           |
| <b>Film Property **</b>                                          |                                         |                   |               |
| Tensile Strength at Yield MD                                     | ASTM D882<br>(500 mm/min)               | MPa               | 10            |
| Tensile Strength at Yield TD                                     |                                         | MPa               | 10            |
| Tensile Strength at Break MD                                     |                                         | MPa               | 25            |
| Tensile Strength at Break TD                                     |                                         | MPa               | 16            |
| Elongation at Break MD                                           |                                         | %                 | 650           |
| Elongation at Break TD                                           |                                         | %                 | 750           |
| Elmendorf Tear Strength MD                                       | ASTM D1922                              | g/μ               | 6             |
| Elmendorf Tear Strength TD                                       |                                         | g/μ               | 16            |
| Dart Impact Strength, F <sub>50</sub> (38 mm Dart, 66 cm Height) | ASTM D1709A                             | g/μ               | 4.0           |
| Gloss at 60°                                                     | ASTM D2457                              | GU                | 76            |
| <b>Suggested Processing Conditions</b>                           |                                         |                   |               |
| Barrel Temperature                                               | 180 – 200 °C                            |                   |               |
| Melt Temperature                                                 | 190 – 210 °C                            |                   |               |
| Blow up Ratio                                                    | 2.0 – 3.0                               |                   |               |
| Die Gap                                                          | 2.0 – 2.5 mm                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

\*\* Typical values with 25  $\mu$  film made with 2.5 mm die gap & BUR 2.5

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 71602S

**71602S** is a LLDPE blown film grade produced by the Spherilene Technology

This grade can be used in General Purpose Packaging Film Application

The precise morphological control achieved during polymerization confers excellent processability & mechanical strengths

**71602S** is additivated with slip & antiblocking agent for good openability & slip properties.

**BIS Designation Code:** IS 7328-2B-FBH-BDB

| Property                                                         | Test Method                             | Unit              | Nominal Value |
|------------------------------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)                                 | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 2.0           |
| Melt Flow Index (5 kg, 190°C)                                    |                                         |                   | 6.0           |
| Melt Flow Index (21.6 kg, 190°C)                                 | ASTM D1238                              | g/10 min          | 60            |
| Density (23°C)                                                   | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.920         |
| DSC Melting Temperature                                          | ASTM D3418                              | °C                | 125           |
| <b>Film Property **</b>                                          |                                         |                   |               |
| Tensile Strength at Yield MD                                     | ASTM D882<br>(500 mm/min)               | MPa               | 11            |
| Tensile Strength at Yield TD                                     |                                         | MPa               | 10            |
| Tensile Strength at Break MD                                     |                                         | MPa               | 19            |
| Tensile Strength at Break TD                                     |                                         | MPa               | 14            |
| Elongation at Break MD                                           |                                         | %                 | 700           |
| Elongation at Break TD                                           |                                         | %                 | 750           |
| Elmendorf Tear Strength MD                                       | ASTM D1922                              | g/μ               | 4             |
| Elmendorf Tear Strength TD                                       |                                         | g/μ               | 12            |
| Dart Impact Strength, F <sub>50</sub> (38 mm Dart, 66 cm Height) | ASTM D1709A                             | g/μ               | 3             |
| <b>Suggested Processing Conditions</b>                           |                                         |                   |               |
| Barrel Temperature                                               | 170 – 200 °C                            |                   |               |
| Melt Temperature                                                 | 190 – 200 °C                            |                   |               |
| Blow up Ratio                                                    | 2.0 – 3.0                               |                   |               |
| Die Gap                                                          | 2.0 – 2.5 mm                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

\*\* Typical values with 25  $\mu$  film made with 2.5 mm die gap & BUR 2.5

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 71602W

**71602W** is a LLDPE film grade produced by the Spherilene Technology

**71602W** is without slip additives and antiblock & can be used in General Purpose Packaging Film Application

The precise morphological control achieved during polymerization confers good processability & mechanical strengths.

**BIS Designation Code:** IS 7328-2B-FBE-BDB

| Property                                                         | Test Method                             | Unit              | Nominal Value |
|------------------------------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)                                 | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 2.5           |
| Melt Flow Index (5 kg, 190°C)                                    |                                         |                   | 6.5           |
| Melt Flow Index (21.6 kg, 190°C)                                 | ASTM D1238                              | g/10 min          | 60            |
| Density (23°C)                                                   | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.920         |
| DSC Melting Temperature                                          | ASTM D3418                              | °C                | 125           |
| <b>Film Property **</b>                                          |                                         |                   |               |
| Tensile Strength at Yield MD                                     | ASTM D882<br>(500 mm/min)               | MPa               | 12            |
| Tensile Strength at Yield TD                                     |                                         | MPa               | 11            |
| Tensile Strength at Break MD                                     |                                         | MPa               | 23            |
| Tensile Strength at Break TD                                     |                                         | MPa               | 19            |
| Elongation at Break MD                                           |                                         | %                 | 500           |
| Elongation at Break TD                                           |                                         | %                 | 750           |
| Elmendorf Tear Strength MD                                       | ASTM D1922                              | g/μ               | 2.5           |
| Elmendorf Tear Strength TD                                       |                                         | g/μ               | 11            |
| Dart Impact Strength, F <sub>50</sub> (38 mm Dart, 66 cm Height) | ASTM D1709A                             | g/μ               | 2             |
| <b>Suggested Processing Conditions</b>                           |                                         |                   |               |
| Barrel Temperature                                               | 170 – 200 °C                            |                   |               |
| Melt Temperature                                                 | 190 – 200 °C                            |                   |               |
| Blow up Ratio                                                    | 2.0 – 3.0 °C                            |                   |               |
| Die Gap                                                          | 2.0 – 2.5 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

\*\* Typical values with 25 μ film made with 2.5 mm die gap & BUR 2.5

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 72307E

**72307E** is a LLDPE produced by the Spherilene Technology

**72307E** can be used in Extrusion Coating on HDPE Woven Fabric, Aluminium Foil, Jute, Paper & Allied Substrates

The precise morphological control achieved during polymerization confers superior processability and improved performance properties not available in conventional LLDPE.

**BIS Designation Code:** IS 7328-2B-HB-CDD

| Property                               | Test Method                             | Unit              | Nominal Value |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 7.0           |
| Melt Flow Index (5 kg, 190°C)          |                                         |                   | 19.0          |
| Melt Flow Index (21.6 kg, 190°C)       | ASTM D1238                              | g/10 min          | 165           |
| Density (23°C)                         | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.926         |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Break              | ASTM D638<br>(50 mm/min)                | MPa               | 18            |
| Elongation at Break                    |                                         | %                 | 500           |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 500           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 300           |
| Vicat Softening Point (10 N)           | ASTM D1525                              | °C                | 102           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 126           |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Barrel Temperature                     | 200 – 270 °C                            |                   |               |
| Die Temperature                        | 250 – 280 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

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## TECHNICAL DATA SHEET

## 73005T

**73005T** is a LLDPE Roto Molding Grade produced by the Spherilene Technology with controlled morphology, excellent balance of processability, mechanical properties.

**73005T** is ideally suited for Tanks & Containers for domestic & industrial water storage, sanitation & agriculture.

**BIS Designation Code:** IS 7328-2B-RBJ-CDC

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 5.0           |
| Melt Flow Index (5 kg, 190°C)                 |                                         |                   | 14            |
| Melt Flow Index (21.6 kg, 190°C)              | ASTM D1238                              | g/10 min          | 120           |
| Density (23°C)                                | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.934         |
| Density (23°C)                                | IS 2530                                 | g/cm <sup>3</sup> | 0.932 – 0.936 |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 17            |
| Elongation at Break                           |                                         | %                 | 800           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 350           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 400           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 110           |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 126           |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 100           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Oven Temperature                              | 240 – 280 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

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## TECHNICAL DATA SHEET

## 73005TU

**73005TU** is a LLDPE Roto Molding Grade produced by the Spherilene Technology with controlled morphology, excellent balance of processability, mechanical properties & UV stability.

This grade is stabilized adequately with Light Stabilizer <sup>+</sup> to give a long service life under outdoor exposure.

**73005TU** is ideally suited for Tanks & Containers with good UV stability for domestic & industrial water storage, sanitation & agriculture.

**BIS Designation Code:** IS 7328-2B-RDO-CDC

| Property                                      | Test Method                             | Unit              | Value         |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 5.0           |
| Melt Flow Index (5 kg, 190°C)                 |                                         |                   | 14            |
| Melt Flow Index (21.6 kg, 190°C)              | ASTM D1238                              | g/10 min          | 120           |
| Density (23°C)                                | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.934         |
| Density (23°C)                                | IS 2530                                 | g/cm <sup>3</sup> | 0.932 – 0.936 |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 17            |
| Elongation at Break                           |                                         | %                 | 800           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 350           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 400           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 110           |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 126           |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 100           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Oven Temperature                              | 240 – 280 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens.

<sup>+</sup> The total Light Stabilizer content is ~2000 ppm.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 73204T

**73204T** is a LLDPE Roto Molding Grade produced by the Spherilene Technology with controlled morphology, excellent balance of processability, mechanical properties.

**73204T** is ideally suited for Tanks & Containers for domestic & industrial water storage, sanitation & agriculture.

**BIS Designation Code:** IS 7328-2B-RBO-CDC

| Property                                      | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)              | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 5.0           |
| Melt Flow Index (5 kg, 190°C)                 |                                         |                   | 14            |
| Melt Flow Index (21.6 kg, 190°C)              | ASTM D1238                              | g/10 min          | 120           |
| Density (23°C)                                | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.932         |
| Density (23°C)                                | IS 2530                                 | g/cm <sup>3</sup> | 0.932 – 0.934 |
| <b>Physical Property</b>                      |                                         |                   |               |
| Tensile Strength at Yield                     | ASTM D638<br>(50 mm/min)                | MPa               | 17            |
| Elongation at Break                           |                                         | %                 | 850           |
| Notched Izod Impact Strength (23°C)           | ASTM D256A                              | J/m               | 500           |
| Flexural Modulus                              | ASTM D790A                              | MPa               | 400           |
| Vicat Softening Point (10 N)                  | ASTM D1525                              | °C                | 109           |
| DSC Melting Temperature                       | ASTM D3418                              | °C                | 126           |
| ESCR (F <sub>50</sub> , 10% Igepal soln. v/v) | ASTM D1693B                             | Hr                | 500           |
| <b>Suggested Processing Conditions</b>        |                                         |                   |               |
| Oven Temperature                              | 240 – 280 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

## 73204TU

**73204TU** is a LLDPE Roto Molding Grade produced by the Spherilene Technology with controlled morphology, excellent balance of processability, mechanical properties & UV stability.

**73204TU** grade is stabilized adequately with Light Stabilizer + to give a long service life under outdoor exposure.

**73204TU** is ideally suited for Tanks & Containers with good UV stability for domestic & industrial water storage, sanitation & agriculture.

**BIS Designation Code:** IS 7328-2B-RDO-CDC

| Property                               | Test Method                             | Unit              | Value         |
|----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 190°C)       | ASTM D1238,<br>IS 13360 (Part 4/Sec 1)  | g/10 min          | 5.0           |
| Melt Flow Index (5 kg, 190°C)          |                                         |                   | 14            |
| Melt Flow Index (21.6 kg, 190°C)       | ASTM D1238                              | g/10 min          | 120           |
| Density (23°C)                         | ASTM D1505,<br>IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.932         |
| Density (23°C)                         | IS 2530                                 | g/cm <sup>3</sup> | 0.932 – 0.934 |
| <b>Physical Property</b>               |                                         |                   |               |
| Tensile Strength at Yield              | ASTM D638<br>(50 mm/min)                | MPa               | 17            |
| Elongation at Break                    |                                         | %                 | 850           |
| Notched Izod Impact Strength (23°C)    | ASTM D256A                              | J/m               | 500           |
| Flexural Modulus                       | ASTM D790A                              | MPa               | 400           |
| Vicat Softening Point (10 N)           | ASTM D1525                              | °C                | 109           |
| DSC Melting Temperature                | ASTM D3418                              | °C                | 126           |
| ESCR (F50, 10% Igepal soln. v/v)       | ASTM D1693B                             | Hr                | 500           |
| <b>Suggested Processing Conditions</b> |                                         |                   |               |
| Oven Temperature                       | 240 – 280 °C                            |                   |               |

\* Halene L is the registered trademark of Linear Low-Density Polyethylene of Haldia Petrochemicals Limited

Mechanical Properties are on Injection Molded test specimens.

+ The total Light Stabilizer content is ~2000 ppm.

This grade meets the requirements of:

IS 7328:2020 Specification for Polyethylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals, and drinking water.

*This product is not recommended for manufacturing of Single Use Plastic (SUP) items listed under Plastics Waste Management (PWM) Rule 2016 and its latest amendment*

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## TECHNICAL DATA SHEET

**B202S**

**B202S** is a Polypropylene Random Copolymer, produced by the Spheripol Technology

**B202S** combines exceptional processability with excellent Melt Strength, High Clarity and Gloss

**B202S** is recommended for Blow Molded Bottles & Containers for Medical Application & Transparent Products

**BIS Designation Code:** IS 10951-2-BB-A

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 1.9           |
| Density (23°C)                          | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 27            |
| Tensile Elongation at Yield             |                                         | %                 | 11            |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 900           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 90            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 130           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 68            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 160 -210 °C                             |                   |               |
| Die Temperature                         | 190 - 210 °C                            |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M212S

**M212S** is a Clarified Polypropylene Random Copolymer produced by Spheripol Technology.

**M212S** combines excellent Clarity, Gloss, High Flow, Low Cycle Time and Stiffness – Impact balance.

**M212S** is recommended for high clarity Injection Molded & Injection Stretch Blow Molded (ISBM) Containers, Household items, Injection Syringes

**BIS Designation Code:** IS 10951-2-MBL-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 12            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 28            |
| Tensile Elongation at Yield             |                                      | %                 | 12            |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1050          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 50            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 128           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 75            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 160 -250 °C                          |                   |               |
| Mold Temperature                        | 30 - 60 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M304

**M304** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M304** combines excellent processability with low Cycle Time and excellent Impact – Stiffness balance.

**M304** is recommended for use in Injection Molded Furniture, Automotive Components, Paint Pails, Crates

**BIS Designation Code:** IS 10951-3-MB-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 3.5           |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 25            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1000          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | >150          |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 90            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 260 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M307

**M307** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M307** combines excellent processability with low Cycle Time

**M307** is recommended for General Purpose Injection Molding

**BIS Designation Code:** IS 10951-3-MB-B

| Property                                | Test Method                             | Unit              | Nominal Value |
|-----------------------------------------|-----------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360<br>(Part 4/Sec 1)  | g/10 min          | 7             |
| Density (23°C)                          | ASTM D1505, IS 13360<br>(Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                         |                   |               |
| Tensile Strength at Yield               | ASTM D638<br>(50 mm/min)                | MPa               | 23            |
| Tensile Elongation at Yield             |                                         | %                 | 6             |
| Flexural Modulus                        | ASTM D790A                              | MPa               | 950           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                              | J/m               | 80            |
| Vicat Softening Point (10 N)            | ASTM D1525                              | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                               | °C                | 88            |
| <b>Suggested Processing Conditions</b>  |                                         |                   |               |
| Barrel Temperature                      | 180 – 260 °C                            |                   |               |
| Mold Temperature                        | 30 – 40 °C                              |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M310

**M310** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M310** combines excellent processability with high Flow, low Cycle Time, excellent Thermal Stability, and good Impact – Stiffness balance.

**M310** is recommended for Injection Molded Battery Box and Industrial Items

**BIS Designation Code:** IS 10951-3-JB-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 10            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 24            |
| Tensile Elongation at Yield             |                                      | %                 | 6             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1200          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 90            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 100           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 160 – 250 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M311T

**M311T** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M311T** combines excellent processability with high flow, low cycle time, good Impact – Stiffness balance with high Toughness and good Gloss.

**M311T** is recommended for Injection Molded Automotive Components, Compounds, Molded Luggage & other Industrial Components

**BIS Designation Code:** IS 10951-3-MB-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 10            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 21            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 900           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 500           |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 142           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 85            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 160 – 250 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M312

**M312** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M312** combines excellent processability with high Flow, low Cycle Time, good Impact – Stiffness balance and good Gloss.

**M312** is recommended for Injection Molded Automotive Components, Housewares, Luggage, Paint Pail, Compounds

**BIS Designation Code:** IS 10951-3-MB-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 12            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 23            |
| Tensile Elongation at Yield             |                                      | %                 | 5             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 950           |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 120           |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 100           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 160 – 250 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M320

**M320** is a Heterophasic Polypropylene Impact Copolymer with Nucleating Agent and produced by the Spheripol Technology

**M320** combines excellent processability with low Cycle Time and good Impact – Stiffness balance.

**M320** is recommended for use in Injection Molded Products, Automotive Components, Appliances

**BIS Designation Code:** IS 10951-3-MBV-D

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 20            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 23            |
| Tensile Elongation at Yield             |                                      | %                 | 6             |
| Flexural Modulus (1% Secant)            | ASTM D790A                           | MPa               | 1150          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 100           |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 105           |
| DSC Melting Temperature                 | ASTM D3418                           | °C                | 165           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 260 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M325

**M325** is a Heterophasic Polypropylene Impact Copolymer produced by the Spheripol Technology

**M325** combines excellent processability with high Flow, low Cycle Time, good Stiffness–Impact balance and good Gloss.

**M325** is recommended for use in Injection Molded Furniture, Compounds & Automotive Components, House wares, Appliance Parts, and Extrusion Coating

**BIS Designation Code:** IS 10951-3-ZB-D

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 25            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 25            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1150          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 60            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 140           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 84            |
| DSC Melting Temperature                 | ASTM D3418                           | °C                | 165           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 210 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of HPL

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M330

**M330** is a Heterophasic Polypropylene Impact Copolymer with Nucleating Agent and produced by the Spheripol Technology

**M330** combines excellent processability with low Cycle Time and good Impact – Stiffness balance.

**M330** is recommended for use in Injection Molded Products, Automotive Parts and Compounds, Appliances

**BIS Designation Code:** IS 10951-3-MBV-D

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 30            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 22            |
| Tensile Elongation at Yield             |                                      | %                 | 6             |
| Flexural Modulus (1% Secant)            | ASTM D790A                           | MPa               | 1100          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 100           |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 100           |
| DSC Melting Temperature                 | ASTM D3418                           | °C                | 165           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 260 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**M340**

**M340** is a Heterophasic Polypropylene Impact Copolymer with Nucleating Agent and produced by the Spheripol Technology

**M340** combines excellent processability with low Cycle Time and good Impact – Stiffness balance.

**M340** is recommended for use in Injection Molded Products, Automotive Parts and Compounds, Appliances

**BIS Designation Code:** IS 10951-3-MBV-E

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 40            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 22            |
| Tensile Elongation at Yield             |                                      | %                 | 6             |
| Flexural Modulus (1% Secant)            | ASTM D790A                           | MPa               | 1050          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 90            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 100           |
| DSC Melting Temperature                 | ASTM D3418                           | °C                | 165           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 260 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**M365**

**M365** is a Heterophasic Polypropylene Impact Copolymer with Nucleating Agent and produced by the Spheripol Technology

**M365** combines excellent processability with low Cycle Time and good Impact – Stiffness balance.

**M365** is recommended for use in Thin-Walled Injection Molded Products, Automotive Parts, Compounds and Appliances.

**BIS Designation Code:** IS 10951-3-MBV-F

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 65            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 22            |
| Tensile Elongation at Yield             |                                      | %                 | 6             |
| Flexural Modulus (1% Secant)            | ASTM D790A                           | MPa               | 1200          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 75            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 148           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 95            |
| DSC Melting Temperature                 | ASTM D3418                           | °C                | 165           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 260 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M308S

**M308S** is a high impact hetero-phasic Polypropylene Impact Copolymer produced by Spheripol Technology.

**M308S** combines excellent processability with low Cycle Time, excellent Impact and Stiffness balance.

**M308S** is recommended for injection molded automotive components, Housewares, Luggage, Paint Pail, Compounds.

**BIS Designation Code:** IS 10951-3-MB-B

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 8             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 20            |
| Tensile Elongation at Yield             |                                      | MPa               | 5             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1000          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 500           |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 145           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 90            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 160 - 250 °C                         |                   |               |
| Mold Temperature                        | 30 - 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10146 for use in contact with foodstuffs, pharmaceuticals and drinking water

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## TECHNICAL DATA SHEET

**F103**

**F103** is a Polypropylene Homopolymer, produced by Spheripol Technology

**F103** is designed for stable extrusion on stenter lines with low water carryover & good thickness control

**F103** has excellent Clarity, Gloss, Mechanical Strengths, Impact & Puncture Resistance

**F103** is recommended for BOPP Films in Food Packaging, Lamination & Metallizable applications

**BIS Designation Code:** IS 10951-1-FB-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 3             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 34            |
| Tensile Elongation at Yield             |                                      | %                 | 10            |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1450          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 40            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 152           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 95            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 230 - 280 °C                         |                   |               |
| Water Bath Temperature                  | 10 - 15 °C                           |                   |               |

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Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**F110**

**F110** is a Polypropylene Homopolymer, produced by the Spheripol Technology

**F110** combines exceptional processability, excellent Clarity and Gloss

**F110** contains slip and antiblocking additives for openability and slip properties

**F110** is recommended for Tubular Quenched (TQ) Film for packaging foodstuffs and garments

**BIS Designation Code:** IS 10951-1-NBH-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 11            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 34            |
| Tensile Elongation at Yield             |                                      | %                 | 9             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1400          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 30            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 154           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 99            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 - 230 °C                         |                   |               |
| Die Temperature                         | 190 - 210 °C                         |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**F135**

**F135** is a Controlled Rheology Polypropylene Homopolymer produced by Spheripol Technology

**F135** combines exceptional processability with high melt flow, narrow molecular weight distribution & gas fading resistance

**F135** is recommended for extrusion coating on woven fabrics and other substrates. This grade can be used in spunbond - nonwoven & other extrusion process and is suitable for hygiene products

**BIS Designation Code:** IS 10951-1-HBV-E

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 35            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 35            |
| Tensile Elongation at Yield             |                                      | %                 | 8             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1350          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 45            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 150           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 105           |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 - 220 °C                         |                   |               |
| Die Temperature                         | 215 - 220 °C                         |                   |               |

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Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**M103**

**M103** is a Polypropylene Homopolymer, produced by the Spheripol Technology

**M103** combines easy processability with high Strength

**M103** is recommended for General Purpose Extrusion & Injection Molding

**BIS Designation Code:** IS 10951-1-MB-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 3             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 32            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1300          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 35            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 153           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 93            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 190 – 250 °C                         |                   |               |
| Mold Temperature                        | 40 – 50 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M106

**M106** is a Polypropylene Homopolymer produced by the Spheripol Technology

**M106** combines easy processability with high Strength

**M106** is recommended for General Purpose Extrusion & Injection Molding

**BIS Designation Code:** IS 10951-1-MB-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 6             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 32            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1300          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 35            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 153           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 93            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 - 260 °C                         |                   |               |
| Mold Temperature                        | 30 - 40 °C                           |                   |               |

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Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**M108**

M108 is a Polypropylene Homopolymer, produced by the Spheripol Technology

M108 combines easy processability with high Strength

M108 is recommended for General Purpose Extrusion & Injection Molding

**BIS Designation Code:** IS 10951-1-MB-B

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 8             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 30            |
| Tensile Elongation at Yield             |                                      | %                 | 7             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1300          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 35            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 152           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 94            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 - 260 °C                         |                   |               |
| Mold Temperature                        | 30 - 40 °C                           |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

## M110

**M110** is a Polypropylene Homopolymer, produced by the Spheripol Technology

**M110** combines exceptional processability with Low Cycle Time, High Gloss & Excellent Color

**M110** is recommended for Injection Molded Furniture & Household items

**BIS Designation Code:** IS 10951-1-MB-C

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 11            |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 34            |
| Tensile Elongation at Yield             |                                      | %                 | 9             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1500          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 30            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 154           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 99            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 180 – 280 °C                         |                   |               |
| Mold Temperature                        | 30 – 40 °C                           |                   |               |

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Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**R103**

**R103** is a Polypropylene Homopolymer produced by the Spheripol Technology

**R103** combines exceptional processability with Low Water Carry Over, Excellent Strength & Color

**R103** is recommended for Stretched Tape Raffia for Woven sacks, FIBC, Ropes & Yarns

**BIS Designation Code:** IS 10951-1-TB-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 3             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 34            |
| Tensile Elongation at Yield             |                                      | %                 | 9             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1500          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 40            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 153           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 93            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 220 – 280 °C                         |                   |               |
| Die Temperature                         | 260 – 280 °C                         |                   |               |
| Quench Temperature                      | 30 – 40 °C                           |                   |               |
| Stretching Temperature                  | ~ 160 °C (Oven)                      |                   |               |

\* Halene P is the registered trademark of Polypropylene of Haldia Petrochemicals Limited

Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

IS 10910:1984 on Specification of Polypropylene for its safe use in contact with Foodstuffs, Pharmaceuticals and Drinking Water

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## TECHNICAL DATA SHEET

**T103**

**T103** is a Polypropylene Homopolymer, produced by the Spheripol Technology

**T103** combines exceptional processability with excellent Stiffness, Clarity & Gloss

**T103** is recommended for Thermoforming of Cups, Containers & other miscellaneous disposable items

**BIS Designation Code:** IS 10951-1-DBL-A

| Property                                | Test Method                          | Unit              | Nominal Value |
|-----------------------------------------|--------------------------------------|-------------------|---------------|
| Melt Flow Index (2.16 kg, 230°C)        | ASTM D1238, IS 13360 (Part 4/Sec 1)  | g/10 min          | 4             |
| Density (23°C)                          | ASTM D1505, IS 13360 (Part 3/Sec 11) | g/cm <sup>3</sup> | 0.90          |
| <b>Physical Property</b>                |                                      |                   |               |
| Tensile Strength at Yield               | ASTM D638 (50 mm/min)                | MPa               | 37            |
| Tensile Elongation at Yield             |                                      | %                 | 8             |
| Flexural Modulus                        | ASTM D790A                           | MPa               | 1700          |
| Notched Izod Impact Strength (23°C)     | ASTM D256A                           | J/m               | 60            |
| Vicat Softening Point (10 N)            | ASTM D1525                           | °C                | 154           |
| Heat Deflection Temperature (0.455 MPa) | ASTM D648                            | °C                | 95            |
| <b>Suggested Processing Conditions</b>  |                                      |                   |               |
| Barrel Temperature                      | 190 – 230 °C                         |                   |               |
| Die Temperature                         | 220 – 250 °C                         |                   |               |

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Mechanical properties tested on Injection Molded Test Specimens prepared in accordance with ASTM D4101

This grade meets the requirements of:

IS 10951:2020 Specification for Polypropylene Material for Moulding and Extrusion

IS 16738:2018 Positive List of Constituents for Polypropylene, Polyethylene and their Copolymers for its Safe Use in Contact with Foodstuffs and Pharmaceuticals

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